

Microsoft 365 Exchange Online

Introduction to PowerShell

PowerShell, developed by Microsoft, is a cross-platform tools that integrates command-line shell, a scripting language, and a configuration management framework. It is designed to assist Help Desk Support and Network Administrators to streamline their IT infrastructures. PowerShell is a scripting language commonly used for automating the management of systems.

Microsoft offers online documentation and sample scripts depending on each use-case scenario. For more information about PowerShell, click [here](#). [PowerShell Documentation](#) by Microsoft is the first go-to place for learning more about an introduction to PowerShell commands. [Sample scripts for system administrators](#) provide a wide collection of practical examples.

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Now that organizations are transitioning to the cloud, learning how to use PowerShell for administering Microsoft 365 Exchange Online becomes vital for IT support specialists. This guide will be highlighting the following steps:

- How to connect to Microsoft Exchange Online PowerShell
- How to change and remove mailbox permissions
- How to change and remove calendar permissions

At the end of the section, a Microsoft 365 Exchange Online cheat sheet is available for additional reference.

How to connect to Microsoft Exchange Online PowerShell

Special note:

Proceed to update PowerShell version via this GitHub link:

<https://github.com/PowerShell/PowerShell/releases/download/v7.5.0/PowerShell-7.5.0-win-x64.exe>

To verify PowerShell version, open PowerShell as administrator and type in: **\$PSVersionTable**

Based on Microsoft detailed guide on how to [Connect to Exchange Online PowerShell](#), it is important to install the module (if it is not already installed). To Install the Microsoft 365 Exchange Online PowerShell Module, use the following PowerShell command.

Install-Module PowerShellGet -Force -AllowClobber

Step 1: Load the Exchange Online PowerShell Module

Once the module is installed, proceed to open a PowerShell window and load the module by running the following command:

Import-Module ExchangeOnlineManagement

Step 2: Connect and authenticate

Run this command in PowerShell to connect and authenticate.

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Connect-ExchangeOnline -UserPrincipalName <admin email address>

For PowerShell 7 for **accounts without MFA**, use

Connect-ExchangeOnline -UserPrincipalName navin@contoso.onmicrosoft.com -InlineCredential

To sign in, use a web browser to open the page <https://microsoft.com/devicelogin> and enter the code <XXXXXXXX> to authenticate. Note the <XXXXXXXX> code value. On any other device with a web browser and internet access, open <https://microsoft.com/devicelogin> and enter the <XXXXXXXX> code value from the previous step. Enter your credentials on the resulting pages. In the confirmation prompt, click Continue. The next message should indicate success, and you can close the browser or tab. The command from step 1 continues to connect you to Exchange Online PowerShell.

Step 3: Disconnect when you are finished.

Be sure to disconnect.

Disconnect-ExchangeOnline

To update the Module (Later On)

Update-Module -Name ExchangeOnlineManagement

How to change mailbox permissions

There are 3 main permission types: Full Access, Send As, and Send on Behalf.

Mailbox Permissions Comparison Chart					
Permission Type	Can Read Mail	Can Send Email	Email Sent As	Email Sent on Behalf	Modify Mailbox
Full Access	Yes	Yes (must open mailbox)	No	No	Yes
Send As	No	Yes	Yes	No	No
Send on Behalf	No	Yes	No	Yes	No

Mailbox Folder Permissions

For more information from Microsoft on mailbox folder permissions, via the [Get-MailboxFolderPermission](#) PowerShell command line. First, check mailbox permissions with the following command line:

Get-MailboxFolderPermission -Identity user@domain.com

For more details on adding mailbox folder permissions go to: [Add-MailboxFolderPermission](#)

Full Access (Access the Entire Mailbox)

Add-MailboxPerission -Identity "user@example.com" -User "delegate@example.com" -AccessRights FullAccess -InheritanceType All

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Send As (Send emails as if they were from the mailbox owner)

```
Add-RecipientPermission -Identity "user@example.com" -Trustee "delegate@example.com" -AccessRights SendAs
```

Send on Behalf (Shows "sent on behalf of")

```
Set-Mailbox -Identity "user@example.com" -GrantSendOnBehalfTo "delegate@example.com"
```

How to remove Mailbox Permission

The process to remove Mailbox Permissions is straightforward by using **Remove-MailboxPermission** commands.

Full Access (Access the Entire Mailbox)

```
Remove-MailboxPermission -Identity user@domain.com -User otheruser@domain.com -AccessRights FullAccess -Confirm:$false
```

Send As

```
Remove-RecipientPermission -Identity user@domain.com -Trustee otheruser@domain.com -AccessRights SendAs -Confirm:$false
```

Send on Behalf

```
Set-Mailbox -Identity user@domain.com -GrantSendOnBehalfTo @{Remove="otheruser@domain.com"}
```

How to change calendar permissions

Calendar Permission Levels permission levels are outlined on Microsoft website [here](#). The three main permissions levels are Reviewer, Editor, and Owner.

Calendar Permissions Comparison Chart					
Permission Level	View Items	Create Items	Edit Items	Delete Items	Delegate Control
Reviewer	Yes	No	No	No	No
Editor	Yes	Yes	Yes	Own	No
Owner	Yes	Yes	Yes	All	Yes

First, check calendar permissions set on a mailbox folder, specifically the **Calendar** folder of a mailbox name "**events**" with the following command below. This PowerShell command lists, who has access to the **calendar** of the **events** mailbox and what level of access they have (Reviewer, Editor, Owner, etc).

```
Get-MailboxFolderPermission -Identity events:\calendar
```

Here is a command line to get a user's calendar permission.

```
Get-MailboxFolderPermission -Identity user@domain.com:\Calendar
```

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An example output might look like this image below.

FolderName	User	AccessRights
-----	----	-----
Calendar	Default	{AvailabilityOnly}
Calendar	John.Doe	{Editor}
Calendar	Jane.Smith	{Reviewer}

Example 1: Reviewer (read-only access to calendar)

Add-MailboxFolderPermission -Identity [user@example.com \Calendar](#) -User "delegate@example.com" -
AccessRights Reviewer

Example 2: Editor (Read / Write)

Add-MailboxFolderPermission -Identity [user@example.com \Calendar](#) -User "delegate@example.com" -
AccessRights Editor

Example 3: Owner (Full Control)

Add-MailboxFolderPermission -Identity [user@example.com \Calendar](#) -User "delegate@example.com" -
AccessRights Owner

Example 4: How to Remove Calendar Permissions

Remove-MailboxFolderPermission -Identity user@domain.com:\Calendar -User otheruser@domain.com
How to add another staff member(s) Calendar in Microsoft Outlook 3